STPM 2015 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ *J s,* $c = 3.00 \times 10^{8}$ *m* s^{-1}, $e = 1.60 \times 10^{-19}$ *C,* $N_A = 6.02 \times 10^{23}$ mol^{-1}, *1 year* $= 3.154 \times 10^{7}$ *s.*

Section A [15 marks]

1. A particle performs a simple harmonic motion along y-axis according to the equation $\frac{d^2y}{dt^2} = -ky$.

Which statement is true of the motion of the particle?

A The value of k does not depend on y.
B The period of the motion depends on y.
C The velocity of the particle is directly proportional to y.
D The acceleration of the particle is always directed away from the equilibrium point.

Answer: A

Comparing with $a = -\omega^2 y$, $k = \omega^2$ is a constant of the system and does not depend on y.

Explanation: The period $2\pi/\sqrt{(k)}$ is independent of displacement; the acceleration is always towards the equilibrium point.

2. An object of mass m oscillates at the end of a spring with a spring constant k. If the amplitude is A, what is the maximum speed of the object?

A $A^2\frac{m}{k}$ B $A^2\frac{k}{m}$ C $A\sqrt{\frac{m}{k}}$ D $A\sqrt{\frac{k}{m}}$

Answer: D

$$v_{max} = A\omega = A\sqrt{(\frac{k}{m})}$$

Explanation: For a mass–spring system $\omega = \sqrt{(\frac{k}{m})}$.

3. A wave motion has an equation $y = 0.20 \sin(200\pi t - 4\pi x)$, where x and y are in meters and t is in seconds. Which is the wavelength and the speed of the wave?

	Wavelength/m	*Speed/m* s^{-1}
A	0.01	0.02
B	0.01	50
C	0.50	0.02
D	0.50	50

Answer: D

$$k = 4\pi \Longrightarrow \lambda = \frac{2\pi}{4\pi} = 0.50 \text{ m}; v = \frac{\omega}{k} = \frac{200\pi}{4\pi} = 50 \text{ m s}^{-1}$$

Explanation: Read ω and k directly from the wave equation.

4. When a sound wave travels in a medium, the displacement y varies with time t and position x according to an equation $y = A \sin(\omega t + kx)$. Which equation shows the variation of pressure change p in the medium?

A $p = p_o \sin(\omega t + kx - \frac{\pi}{2})$

B $p = p_o \sin(\omega t + kx + \frac{\pi}{2})$

C $p = p_o \sin(\omega t + kx + \pi)$

D $p = p_o \sin(\omega t + kx - \pi)$

Answer: B

For a sound wave the pressure variation is a quarter of a cycle ($\pi/2$) out of phase with the displacement: the pressure is greatest where the displacement is zero.

Following the standard result that the pressure wave leads the displacement wave by $\pi/2$: $p = p_o \sin(\omega t + kx + \pi/2)$ (B).

Explanation: Strictly, $p = -B\,\partial y/\partial x$; for a displacement written with $+kx$ this gives a phase of $-\pi/2$ (option A). The key answer follows the standard 'pressure leads displacement by $\pi/2$' result.

5. An ambulance and a car move towards each other with a velocity of 20.0 m s^{-1} and 10.0 m s^{-1} respectively. If the frequency of the siren of the ambulance is 500 Hz, which is the relative velocity and apparent frequency of the sound received by the car driver?

[The speed of sound in air is 340 m s^{-1}.]

	Relative velocity of sound/m s^{-1}	*Apparent frequency of sound/Hz*
A	340	458
B	340	547
C	350	458
D	350	547

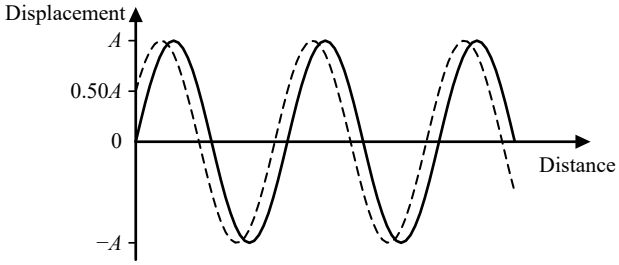
Answer: D

Speed of sound relative to the car = 340 + 10.0 = 350 m s^{-1}

$$f' = 500\frac{340 + 10.0}{340 - 20.0} = 547 \text{ Hz}$$

Explanation: Both the source and the observer move towards each other, so the frequency rises.

6. Two identical progressive waves, each of them with an amplitude A and intensity I_o but of different phases are shown in the graph below.



If the two waves superpose, which is the phase difference and the intensity of the resultant wave?

	Phase difference	*Intensity*
A	$\frac{\pi}{6}$	$2.0I_o$
B	$\frac{\pi}{2}$	$2.0I_o$
C	$\frac{\pi}{6}$	$3.7I_o$
D	$\frac{\pi}{2}$	$3.7I_o$

Answer: C

From the graph, at $x = 0$ one wave is at 0 and the other at $0.50A$: $\sin\phi = 0.50 \Longrightarrow \phi = \pi/6$.

$$\text{Resultant amplitude} = 2A\cos\frac{\pi}{12} = 1.93A \Longrightarrow I = (1.93)^2 I_o = 3.7I_o$$

Explanation: Intensity is proportional to amplitude squared; two waves almost in phase give nearly four times the intensity of one.

7. An object is placed 5.0 cm in front of a concave mirror with radius of curvature 20.0 cm. Which statement is true about the image formed by the mirror?

A It is enlarged, upright and 10 cm behind the mirror.
B It is enlarged, inverted and 10 cm in front of the mirror.
C It is diminished, upright and 6.7 cm behind the mirror.
D It is diminished, inverted and 6.7 cm in front of the mirror.

Answer: A

$$f = 10.0\text{ cm}: \frac{1}{v} = \frac{1}{10.0} - \frac{1}{5.0} \Longrightarrow v = -10\text{ cm};\ m = 2$$

Enlarged, upright, virtual, 10 cm behind the mirror.

Explanation: The object is inside the focal length, so the concave mirror acts as a magnifying (shaving) mirror.

8. Which light source is incoherent?

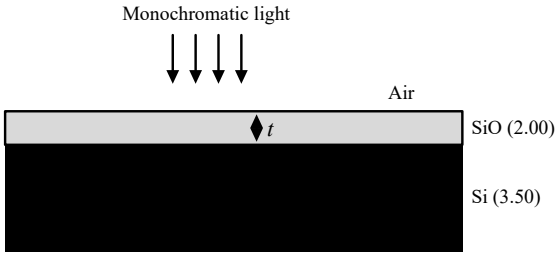
A Light sources from a diffraction grating
B Light of the same frequencies from two separate sources.
C Light that is partially reflected from the front and back layers of thin films
D Light sources from a narrow double-slits after passing through a single slit

Answer: B

Light from two separate sources is incoherent: even with the same frequency, the phase difference between them changes randomly.

Explanation: Coherent sources are obtained by splitting light from one source (slits, grating, thin films).

9. A solar cell consists of silicon (Si) with a coating of silicon oxide (SiO) is shown in the diagram below.



If the refractive indices of silicon and silicon oxide are 3.50 and 2.00 respectively, what is the minimum thickness t of the coating so that monochromatic light of wavelength 600 nm is not reflected from the surface of the coating?

A 43.0 nm B 75.0 nm C 86.0 nm D 150 nm

Answer: B

Both reflections (air → SiO and SiO → Si) have a phase change of π, which cancel.

$$\text{Destructive: } 2nt = \frac{\lambda}{2} \Longrightarrow t = \frac{600}{4(2.00)} = 75.0 \text{ nm}$$

Explanation: The coating is a quarter of a wavelength thick in the coating.

10. In an experiment on photoelectric effect, a target material is irradiated with a beam of monochromatic light. Which action causes the rate of emission of photoelectrons to increase?

A Intensity of light is increased.
B Intensity of light is decreased.
C Light with lower frequency is used.
D Light with higher frequency is used.

Answer: A

Increasing the intensity increases the number of photons per second, so more photoelectrons are emitted per second.

Explanation: Raising the frequency increases the kinetic energy of the photoelectrons, not their number.

11. Why does an electron microscope have a higher resolving power than an optical microscope?

A Electron travel slower than visible light.
B Electron have charge but visible light is neutral.
C Electrons have shorter wavelength than visible light.
D Electrons have higher penetrating power than visible light.

Answer: C

Fast electrons have a much shorter de Broglie wavelength than visible light, so they are diffracted less and can resolve finer detail.

Explanation: Resolving power is limited by diffraction, which depends on wavelength.

12. Electrons in an X-ray tube are accelerated by a potential difference of 10 kV. What is the minimum wavelength of the resulting X-rays?

A 1.2×10^{-12} m B 1.2×10^{-11} m C 1.2×10^{-10} m D 1.2×10^{-9} m

Answer: C

$$\lambda_{\min} = \frac{hc}{eV} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(1.60 \times 10^{-19})(10 \times 10^{3})} = 1.2 \times 10^{-10}\ \text{m}$$

Explanation: The whole kinetic energy of one electron becomes one photon.

13. The half-life of radium is 1620 years. If the mass of a sample of radium is 1.00 g, what is the number of nuclei that decays in 1 s?
[Molar mass of radium is 226 g mol^{-1}]

A 1.57×10^{10} B 3.61×10^{10} C 3.59×10^{11} D 1.32×10^{13}

Answer: B

$$N = \frac{1.00}{226}(6.02 \times 10^{23}) = 2.66 \times 10^{21};\ \lambda = \frac{\ln 2}{(1620)(3.154 \times 10^{7})} = 1.36 \times 10^{-11}\ \text{s}^{-1}$$

$$A = \lambda N = 3.61 \times 10^{10}\ \text{s}^{-1}$$

Explanation: Convert the half-life to seconds to get the activity per second.

14. What is the binding energy per nucleon of $^{202}_{80}\text{Hg}$ nucleus?
[The atomic mass of $^{202}_{80}\text{Hg} \equiv 201.970\ 617$ u, atomic mass of $^{1}_{1}\text{H} = 1.007\ 825$ u, mass of $^{1}_{0}\text{n} = 1.008\ 665$ u and 1 u ≡ 934 MeV.]

A 7.76 MeV B 7.92 MeV C 13.1 MeV D 20.0 MeV

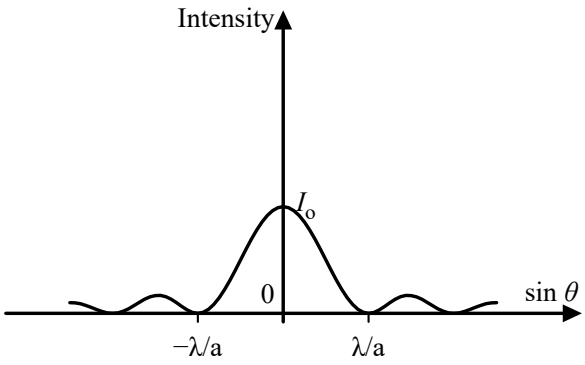
Answer: B

$$\Delta m = 80(1.007825) + 122(1.008665) - 201.970617 = 1.712513 \text{ u}$$

$$E_B/A = \frac{(1.712513)(934)}{202} = 7.92 \text{ MeV}$$

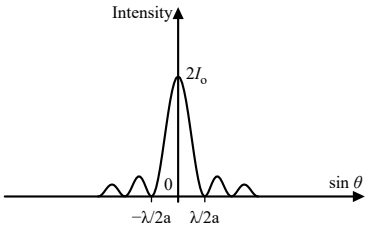
Explanation: Mercury-202 has 80 protons and 122 neutrons. Use 1 u ≡ 934 MeV as given.

15. The intensity distribution of a diffraction pattern produced on a screen when a monochromatic light with wavelength λ is incident on a single slit of width a is shown in the graph below.

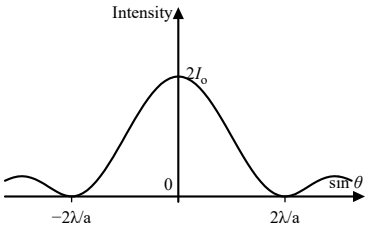


Which graph represents the intensity distribution when the slit is replaced with a slit of width $2a$?

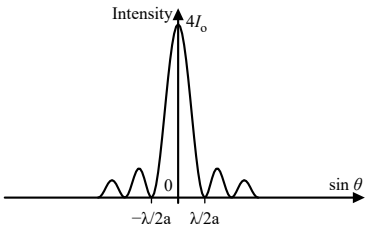
A



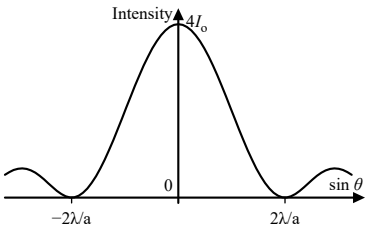
B



C



D



Answer: C

Doubling the slit width halves the width of the central maximum (minima at $\pm\lambda/2a$).

Twice as much light passes and the amplitude at the centre doubles, so the central intensity becomes $4I_o$ (C).

Explanation: The central intensity is proportional to the square of the slit width.

Section B [15 marks]

16. (a) Two flutes produce sounds with frequencies 259 Hz and 261 Hz at the same time. Determine the beat frequency. [2]

Answer

Beat frequency = $|f_1 - f_2| = 261 - 259$ — 1

= 2 Hz — 1

2

Explanation: The loudness rises and falls twice per second.

(b) A guitar string of length 63.5 cm is tuned to produce a fundamental note of 245 Hz.

(i) Calculate the speed of the wave on the string. [3]

Answer

Fundamental: $L = \frac{\lambda}{2} \Longrightarrow \lambda = 2(0.635) = 1.27$ m — 1

$v = f\lambda = (245)(1.27)$ — 1

$v = 311$ m s^{-1} — 1

3

Explanation: This is the speed of the transverse wave on the string, not the speed of sound in air.

(ii) Explain the change in frequency of the wave on the string if the tension of the string is increased. [2]

Answer

Increasing the tension increases the wave speed ($v = \sqrt{(\frac{T}{\mu})}$) — 1

The length (wavelength) is fixed, so the frequency increases ($f = v/2L$) — 1

2

Total for question **7**

Explanation: This is how a guitar is tuned.

17. (a) State the principle that explains diffraction. [1]

Answer

Huygens's principle: every point on a wavefront acts as a source of secondary wavelets, and the new wavefront is the envelope of these wavelets — 1

1

Explanation: At the edges of an obstacle or slit, the wavelets spread into the shadow region.

(b) A monochromatic light with frequency 6.0×10^{14} Hz is incident normally on a diffraction grating which has 300 lines per mm.

(i) Determine the angle of diffraction for second order maximum. [4]

Answer

$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{6.0 \times 10^{14}} = 5.0 \times 10^{-7}$ m — 1

$d = \frac{1}{300 \times 10^3} = 3.33 \times 10^{-6}$ m — 1

$\sin\theta = \frac{2\lambda}{d} = \frac{2(5.0 \times 10^{-7})}{3.33 \times 10^{-6}} = 0.30$ — 1

$\theta = 17.5°$ — 1

4

Explanation: 300 lines per mm = 3.00×10^5 lines per metre.

(ii) Determine the number of maximum produced. [3]

Answer

$n \leq \frac{d}{\lambda} = \frac{3.33 \times 10^{-6}}{5.0 \times 10^{-7}} = 6.67$ — 1

Highest order $n = 6$ — 1

Number of maxima = 2(6) + 1 = 13 — 1

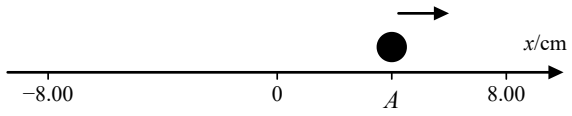
3

Total for question **8**

Explanation: Count the zero order and the orders on both sides.

Section C [30 marks]

18. (a) An object is projected from point A and starts to oscillate in a simple harmonic motion with an amplitude of 8.00 cm as shown in the diagram below.



At point A the object is at half of its amplitude. Its angular frequency is 20.0 rad s^{-1}. Derive an expression that represents the motion of the object. [3]

Answer

$x = A \sin(\omega t + \phi) = 8.00 \sin(20.0t + \phi)$ cm — 1

At $t = 0$, $x = 4.00$ cm: $\sin \phi = 0.50$, moving in the $+x$ direction $\Longrightarrow \phi = \frac{\pi}{6}$ — 1

$x = 8.00 \sin(20.0t + \frac{\pi}{6})$ cm — 1

3

Explanation: The other solution, $\phi = 5\pi/6$, would mean the object is moving in the negative direction at A.

(b) A mass of 0.750 kg is attached to a spring with a spring constant of 350 N m^{-1}. The system executes a simple harmonic motion horizontally on a smooth surface with an amplitude of 6.00 cm. Calculate

(i) the period of the motion, [3]

Answer

$T = 2\pi\sqrt{(\frac{m}{k})}$ — 1

$= 2\pi\sqrt{(\frac{0.750}{350})}$ — 1

$T = 0.291$ s — 1

3

Explanation: The period does not depend on the amplitude.

(ii) the speed of the mass at 2.50 cm from the equilibrium, [2]

Answer

$v = \omega\sqrt{(A^2 - x^2)} = \sqrt{(\frac{350}{0.750})}\sqrt{(0.0600^2 - 0.0250^2)}$ — 1

$v = 1.18$ m s^{-1} — 1

2

Explanation: $\omega = 21.6$ rad s^{-1}.

(iii) the kinetic energy of the mass at 3.50 cm from the equilibrium. [2]

Answer

$K = \frac{1}{2}k(A^2 - x^2) = \frac{1}{2}(350)(0.0600^2 - 0.0350^2)$ 1

$K = 0.416$ J 1

2

Explanation: Kinetic energy = total energy − potential energy.

(c) If the horizontal surface in (b) is not smooth, describe the changes in the motion of the mass. [3]

Answer

Friction causes damping: energy is lost as heat 1

The amplitude decreases gradually with time until the mass stops 1

The oscillation is no longer simple harmonic (the period increases slightly) 1

3

Explanation: Light damping gives underdamped oscillations.

(d) State whether it is possible for damped oscillation to occur while a system is in resonance condition. Give your reason. [2]

Answer

Yes 1

Resonance occurs in a forced, damped system when the driving frequency equals the natural frequency; the driving force replaces the energy lost by damping, and damping limits the resonance amplitude 1

2

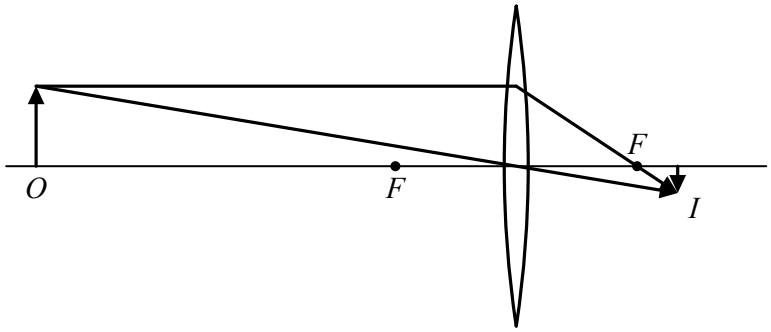
Total for question **15**

Explanation: Heavier damping gives a lower, broader resonance peak.

19. (a) An object is placed 30.0 cm in front of a converging lens with a focal length of 7.55 cm.

(i) Sketch the ray diagram of the image. [2]

Answer



Two correct rays from the top of the object (parallel ray through F; ray through the optical centre)	1
Real, inverted image formed beyond F on the other side	1
	2

Explanation: The object is beyond $2F$ (15.1 cm), so the image lies between F and $2F$.

(ii) Calculate the position of the image. [2]

Answer

$\frac{1}{v} = \frac{1}{7.55} - \frac{1}{30.0}$	1
$v = 10.1$ cm (behind the lens)	1
	2

Explanation: Positive v means a real image on the opposite side of the lens.

(iii) Determine the magnification of the image. [2]

Answer

$m = \frac{v}{u} = \frac{10.1}{30.0}$	1
$m = 0.337$	1
	2

Explanation: The image is about a third of the object height.

(iv) State the characteristic of the image. [1]

Answer

Real, inverted and diminished — 1

1

Explanation: As expected for an object beyond 2*F*.

(b) A 10.0 cm tall object is placed 15.0 cm in front of a convex mirror with a radius of curvature 40.0 cm. Calculate

(i) the position of the image, [3]

Answer

$f = -\frac{40.0}{2} = -20.0$ cm — 1

$\frac{1}{v} = \frac{1}{-20.0} - \frac{1}{15.0}$ — 1

$v = -8.57$ cm (virtual, behind the mirror) — 1

3

Explanation: A convex mirror always gives a virtual image behind it.

(ii) the height of the image. [2]

Answer

Height $= \frac{8.57}{15.0} \times 10.0$ — 1

$= 5.71$ cm — 1

2

Explanation: Upright and diminished.

(c) A spider with a diameter of 2.00 cm hangs from the ceiling. The distance between the spider and the wall is 200 cm. A thin lens is placed so that an image formed on the wall is half the size of the spider.

(i) State the type of lens used. [1]

Answer

A converging (convex) lens — 1

1

Explanation: Only a converging lens can form a real image on a screen (the wall).

(ii) Determine the position of the lens from the wall. [2]

Answer

$u + v = 200$ cm and $v = \frac{u}{2}$ 1

$u = 133$ cm, $v = 66.7$ cm: the lens is 66.7 cm from the wall 1

2

Total for question **15**

Explanation: A magnification of ½ means the image distance is half the object distance.

20. (a)

(i) Describe the Bohr atomic model. [2]

Answer

Electrons move in certain circular orbits (stationary states) around the nucleus without radiating; the angular momentum is quantised, $mvr = nh/2\pi$ 1

Radiation is emitted or absorbed only when an electron moves between orbits: $hf = E_i - E_f$ 1

2

Explanation: The Coulomb force provides the centripetal force in each orbit.

(ii) State the weaknesses of the Bohr atomic model for atoms other than hydrogen-like atoms. [2]

Answer

It cannot explain the spectra of multi-electron atoms (it ignores electron–electron interactions) 1

It cannot explain the relative intensities or the fine structure of spectral lines (and conflicts with the uncertainty principle) 1

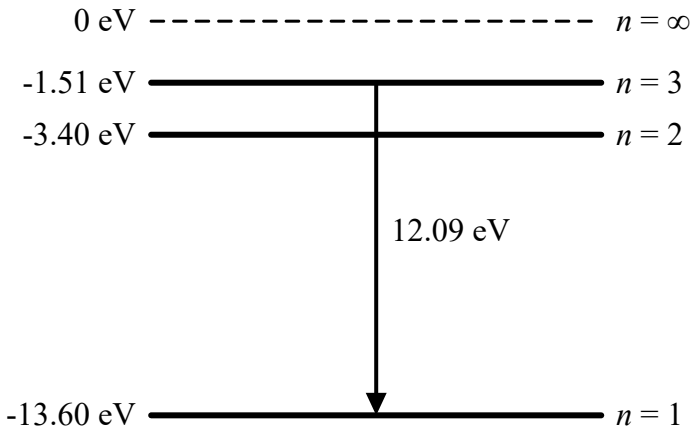
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Explanation: It works only for one-electron (hydrogen-like) systems.

(b) A hydrogen atom at its nth state has energy E_n is given by the expression $E_n = (-$ ${frac('13.6','n^2')}) eV.

(i) Sketch and label the energy-level diagram for the three lowest energy states of the hydrogen atom. [3]

Answer



Three levels drawn with energies −13.6 eV ($n = 1$), −3.40 eV ($n = 2$), −1.51 eV ($n = 3$)	2
Levels closer together at higher n; zero level ($n = \infty$) marked	1
	3

Explanation: $E_n = -13.6/n^2$ eV.

(ii) Determine the wavelength of photon emitted when an excited hydrogen atom at $n = 3$ returns to the ground state. [4]

Answer

$\Delta E = -1.51 - (-13.6) = 12.09$ eV	1
$\Delta E = (12.09)(1.60 \times 10^{-19}) = 1.93 \times 10^{-18}$ J	1
$\lambda = \frac{hc}{\Delta E} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{1.93 \times 10^{-18}}$	1
$\lambda = 1.03 \times 10^{-7}$ m	1
	4

Explanation: This is an ultraviolet line of the Lyman series.

(iii) Calculate the wavelength of a photon which is able to ionise the hydrogen atom at ground state. [2]

Answer

Ionisation from $n = 1$ needs 13.6 eV: $\lambda = \dfrac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(13.6)(1.60 \times 10^{-19})}$ 1

$\lambda = 9.14 \times 10^{-8}$ m 1

2

Explanation: Photons of shorter wavelength also ionise the atom, giving the electron extra kinetic energy.

(c) State **two** series of line spectrums obtainable from the hydrogen atom and state which one is visible. [2]

Answer

Lyman series (transitions to $n = 1$) – ultraviolet 1

Balmer series (transitions to $n = 2$) – visible 1

2

Total for question **15**

Explanation: The Paschen series (to $n = 3$) is in the infrared.